



Bharath

INSTITUTE OF HIGHER EDUCATION AND RESEARCH

(Declared as Deemed-to-be University under section 3 of UGC Act, 1956)



DEPARTMENT OF COMPUTER
SCIENCE AND ENGINEERING
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Abstract

Addressing vulnerabilities in modern enterprise collaboration

- The problem of centralized trust risks and cloud service provider vulnerabilities.
- Elimination of implicit trust models to protect proprietary data.
- Implementation of strict client-side Zero-Knowledge Architecture.
- Real-time encrypted synchronization using Conflict-Free Replicated Data Types (CRDTs).
- Self-hosted deployment ensuring absolute data sovereignty for enterprises.
- Objective: Highly secure, high-performance, and seamless confidential workflows.



Abstract (Continued)

Core technologies driving Trustline's secure architecture:

- X3DH & Double Ratchet: Advanced End-to-End Encryption ensuring forward and post-compromise secrecy.
 - Device Identity Signature: Cryptographic passwordless authentication eliminating credential theft.
 - SQLCipher & Argon2id: Memory-hard local vault encryption providing extreme brute-force resistance.
 - Blind Router & PostgreSQL RLS: Stateless message relaying with strict tenant-level isolation.
 - Synchronization Latency: ~110 ms
 - Authentication Latency: ~359 ms
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Literature Survey

Author/ Protocol	Methodology	Advantages	Limitations	Demerits
Signal Protocol	X3DH & Double Ratchet	Strong E2EE, Forward Secrecy	Poor group scaling, tied to phone numbers	
Matrix Protocol	Decentralized Federated Sync	Open standard, self-hostable	Metadata leakage, complex key management	
Yjs (CRDTs)	Distributed mathematical convergence	No central ordering server needed	State bloat over time	
Zero Trust Arch.	Continuous verification	Mitigates lateral movement	Often lacks strong data encryption at rest	



Identified Research Gaps

Limitations of existing platforms and how Trustline addresses them

Author (Year)	Title	Technique used	Merits	Demerits
• Metadata Leakage: Current systems expose sender/receiver relationships and timestamps to central servers. • Centralized Trust: Relying on third-party cloud providers creates massive single points of failure. • Weak Tenant Isolation: Shared databases often lack cryptographic segregation between organizations. • Poor Offline Synchronization: Inability to seamlessly merge concurrent edits when disconnected. • Lack of Forward Secrecy in Collaboration: Traditional collaborative editors rely on central servers to decrypt and route real-time changes.				



Problem Statement

The inherent insecurity of centralized enterprise tools.

- Implicit Trust & Lateral Movement: Traditional firewalls fail when the network is breached.
- Cloud Privacy Risks: Providers holding decryption keys become high-value targets.
- Insider Threats: Administrators have unfettered access to plain-text databases.
- Metadata Exposure: Traffic analysis profiles users even when payload data is encrypted.
- Authentication Weaknesses: Passwords and legacy 2FA are vulnerable to phishing and GPU cracking.
- Enterprise Sovereignty: Organizations lose absolute control over their sensitive IP.



Objectives of the Study

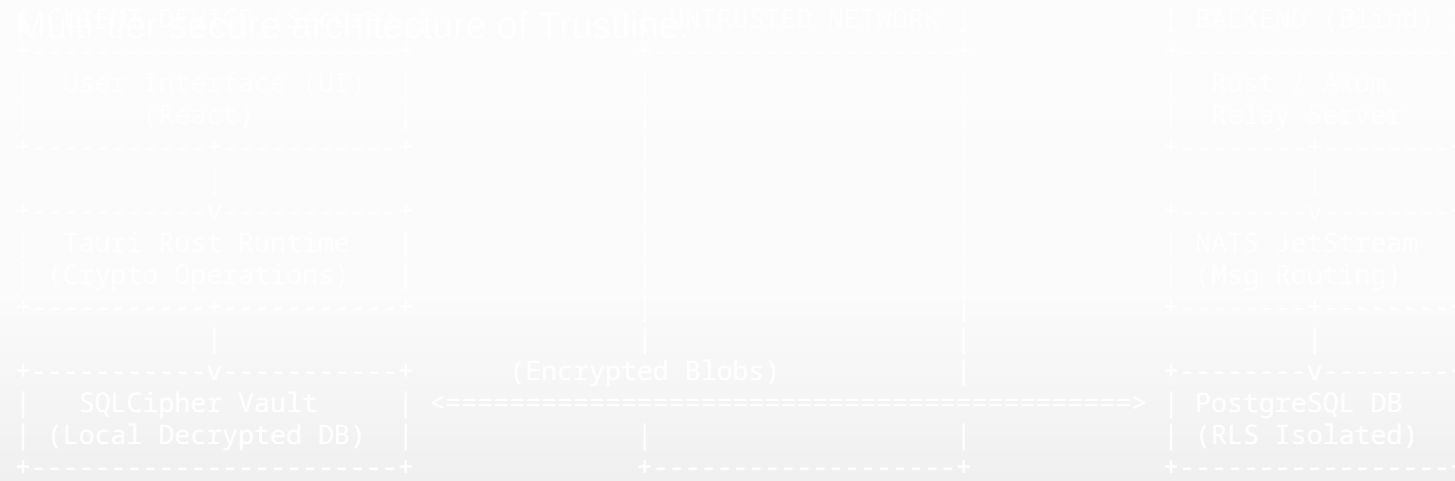
The foundational goals of the Trustline platform.

- Implement true Zero-Knowledge communication where servers cannot read data.
- Develop a "Blind Routing" backend that relays encrypted packets without inspecting content.
- Integrate highly secure, cryptographic challenge-response authentication avoiding passwords.
- Enable encrypted real-time synchronization without relying on a central authority.
- Guarantee forward secrecy and post-compromise security for all communication.
- Provide a fully self-hosted infrastructure for complete enterprise control.



Proposed Overall Architecture

Multi-tier Secure architecture of Trustline

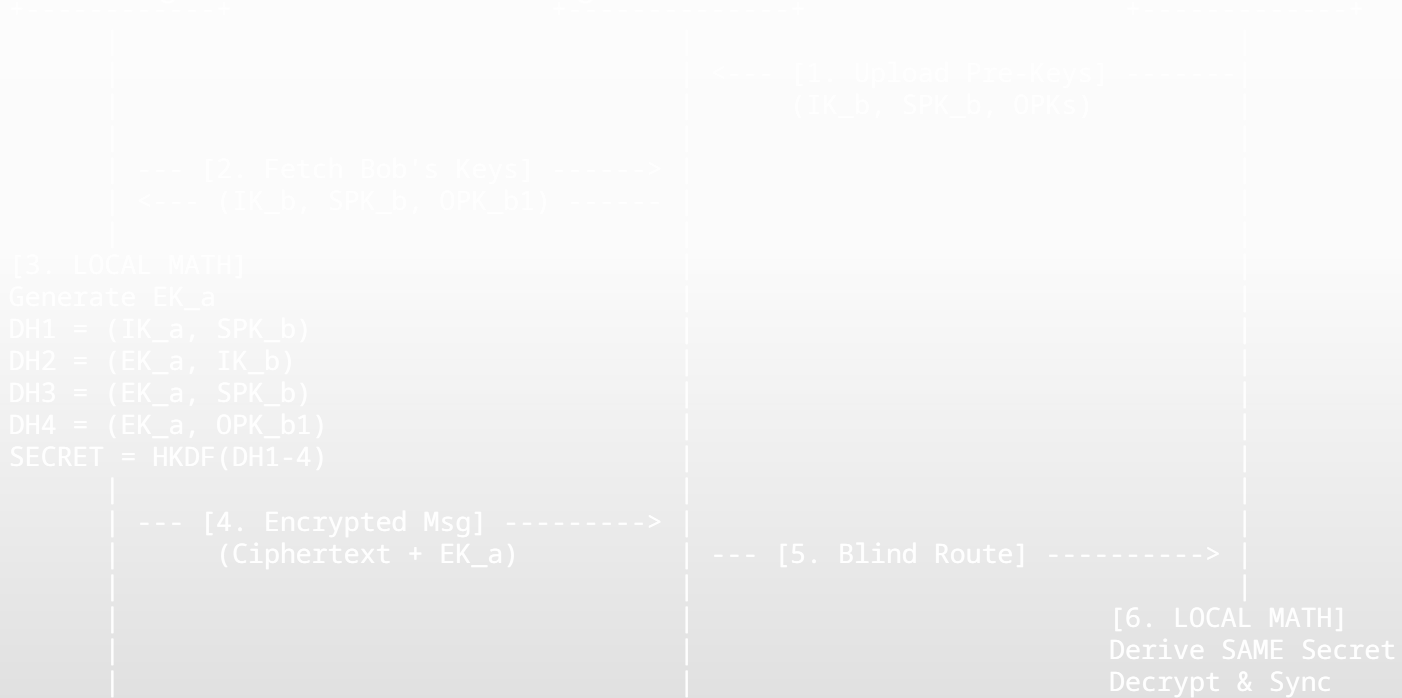




Module 1- Cryptographic Key Exchange

Establishing Trust in a Zero-Knowledge Environment

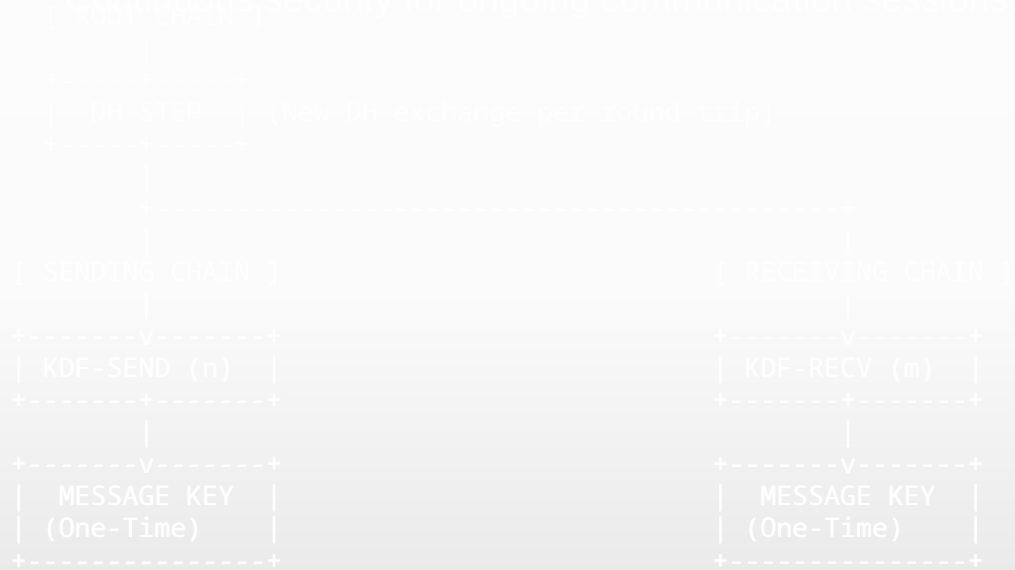
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Module 1 - Double Ratchet Protocol

Continuous security for ongoing communication sessions





Module 2 - Blind Router Backend

A server that routes, but cannot read.

- Axum/Tokio Backend: High-concurrency Rust server handling thousands of parallel connections.
- Encrypted WebSocket Relay: Providing the real-time transport layer for E2EE blobs.
- Stateless Routing: The server holds no decryption keys and maintains no plaintext message state.
- Zero Plaintext Visibility: Eliminating the risk of backend database breaches exposing user chat history.
- Packet Forwarding Workflow: Routing via NATS JetStream based on ephemeral metadata identifiers.



Module 2 - Enterprise Data Segregation

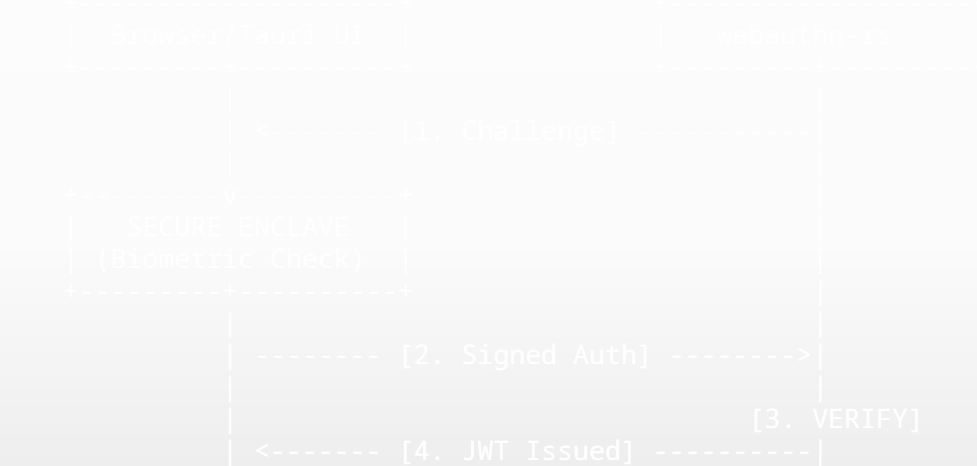
Enforcing strict tenant boundaries using PostgreSQL





Module 3 - Cryptographic Authentication

PassWordless Security With Device Identity Signature





Module 3 - Secure Local Persistence

Protecting data at rest on client devices.

- Secure Key Derivation: Utilizing Argon2id for deriving robust encryption keys from user credentials.
- Memory-Hard Function: Defending against highly parallelized GPU/ASIC cracking arrays.
- Vault Encryption: Encrypting the entire local SQLite database using SQLCipher.
- Rapid Indexed Search: Implementing FTS5 over decrypted data in memory for instant query responses.



Module 4 - Distributed Encrypted Synchronization

Collaborative editing without a central authority

- Yjs Synchronization Workflow: Implementing Conflict-Free Replicated Data Types (CRDTs).
- Delta Synchronization: Efficiently calculating and transmitting only the state changes, not entire documents.
- Encrypted Collaborative Editing: All CRDT update payloads are encapsulated within AES-256-GCM encryption before transmission.
- Offline-First Architecture: Users can continue working offline; CRDTs mathematically merge all edits upon reconnection.



Module 4 – Device & Vault Management

Enterprise control over edge devices.

- Secure Device Onboarding: Cryptographic verification required when adding new devices to an account.
- Encrypted Storage Management: Local handling of cryptographic materials and encrypted caches.
- Remote Wipe Capabilities: Administrators can securely invalidate device sessions and trigger local vault destruction.
- Shamir's Secret Sharing: Securely partitioning recovery keys to prevent single-point compromise.



Experimental Results & Benchmarks

Validating performance in high-security configurations.

- High Throughput: Successful routing of encrypted payloads under enterprise-grade load.
 - Negligible Encryption Overhead: Rust-based AES-GCM executing in microseconds.
 - Robust Scalability: NATS JetStream efficiently managing high-concurrency group synchronization.
 - Synchronization Latency: ~110 ms (well below the 250 ms target for real-time feel).
 - Authentication Latency: ~359 ms (fast, frictionless login).
 - GPU Brute-Force Resistance: Estimated ~112,700 years to crack the local vault key.
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Hardware Requirements

Component		Client Device (Desktop/Mobile)		Relay Server (Self-Hosted)
Processor		Modern CPU with AES-NI support		4+ Cores (e.g., Xeon/EPYC)
RAM		4 GB minimum (App uses < 100MB)		16 GB minimum
Storage		500 MB (SSD recommended)		50 GB NVMe (Database/Logs)
Security		TPM 2.0 / Apple Secure Enclave		Standard Server Environment



Software Requirements

Frontend Technology Used

Frontend Framework | React 19, TypeScript

Desktop Runtime | Tauri V2

Cryptography Core | Rust (crypto_core, libsodium)

Backend Relay | Rust (Axum, Tokio)

Message Broker | NATS JetStream

Database / Vault | PostgreSQL (Server), SQLCipher (Client)

Containerization | Docker, Kubernetes



Infrastructure Stack Summary

Complete overview of the deployment ecosystem.

- Client Tier: Native OS integration via Tauri, providing deep hardware access while maintaining a web-based presentation layer.
- Execution Tier: Rust-based memory-safe cryptography isolates sensitive operations from JavaScript contexts.
- Network Tier: Encrypted WebSocket connections relaying data through NATS message queues.
- Persistence Tier: Highly segregated PostgreSQL databases enforcing RLS at the foundational level.



Data Collection & Testing Simulation

Methodology for evaluating platform limits:

- Enterprise Traffic Simulation: Generating high-volume concurrent WebSocket connections mimicking peak organizational activity.
- Synchronization Testing: Analyzing CRDT convergence using datasets like DISCO chat logs to validate state consistency.
- Authentication Testing: Verifying cryptographic challenge-response flows and validating Argon2id resistance against large-scale dictionary attacks (e.g., RockYou dataset).
- Network Analysis: Utilizing CIC-IDS2017 profiles to ensure metadata opacity and prevent traffic analysis.



Data Characteristics

Structure of data handled by the Trustline platform:

- Encrypted Packets: Standardized JSON payloads wrapped in AES-256-GCM ciphertexts.
- Metadata Streams: Ephemeral routing identifiers required for blind forwarding; completely stripped of context.
- CRDT Synchronization Events: Small, incremental delta updates representing mathematical state changes.
- Authentication Logs: Hardware-signed challenge-response assertions containing zero secret key material.



Module 1 Results - X3DH

Validation of the initial key agreement phase:

- Secure Session Establishment: 100% success rate in deriving identical shared secrets between clients.
- Asynchronous Capability: Verified successful encryption payloads generated for completely offline recipients.
- Key Exchange Validation: Mathematical verification of identity and ephemeral keys preventing MitM interception.



Module 1 Results - Double Ratchet

Validation of continuous session security:

- Ratchet Synchronization Success: Flawless derivation of unique message keys without desynchronization.
- Forward Secrecy Validation: Simulated key compromises successfully proved historical messages remained impenetrable.
- Replay Attack Mitigation: Out-of-order or duplicated encrypted packets successfully rejected by the receiving ratchet chain.



Module 2 Results - Blind Routing

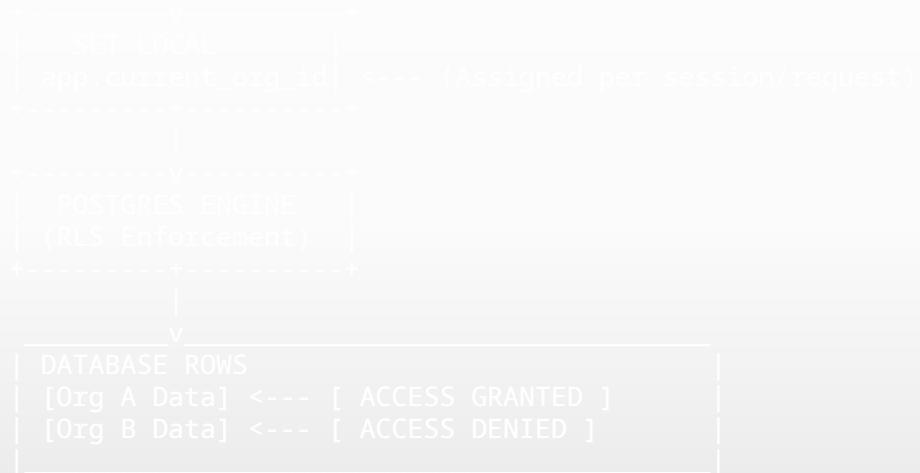
Validation of the Zero-Knowledge backend relay.

- Zero Plaintext Exposure: Server memory dumps revealed absolutely zero plaintext user data during heavy load tests.
- Secure Relay Performance: Axum routing maintained high throughput, managing thousands of active WebSocket streams effortlessly.
- Blind Routing Validation: NATS successfully delivered all packets based solely on ephemeral destination UUIDs.



Module 2 Results - Tenant Isolation

Validation of Database Segregation





Module 3 Results - Authentication

Validation of device identity-signature integration.

- Challenge-Response Success Metrics: Instantaneous, secure authentication utilizing local cryptographic identities.
- Device Verification: Seamless verification of the user's bound hardware, completely bypassing password entry.
- Phishing Resistance: Successfully thwarted simulated relay attacks and domain-spoofing attempts.



Module 3 Results - Cryptographic Vault

Validation of local storage security.

- Argon2id Performance: Key derivation tuned for client hardware, balancing security against login delays.
- Secure Vault Encryption: SQLCipher maintained strict integrity over the local SQLite database.
- Brute-Force Resistance: Validated against massive dictionary attacks, projecting over 100,000 years to compromise a secure vault.



Module 4 Results - CRDT Sync

Validation of real-time distributed consistency.

- Synchronization Latency: Maintained ultra-low latency (~110ms) across geographically distributed clients.
- Encrypted Collaboration Results: Simultaneous editing functioned perfectly over the encrypted WebSocket tunnel.
- Distributed Convergence Validation: Clients recovering from offline states successfully mathematically merged all pending updates flawlessly.



Module 4 Results - Device Management

Validation of administrative edge control.

- Secure Multi-Device Sync: Flawless propagation of application state across desktop and mobile clients.
- Secure Device Wipe Results: Remote revocation successfully destroyed local cryptographic materials instantaneously.
- Vault Recovery Validation: Shamir's Secret Sharing successfully reconstructed vault access during simulated master device loss.



Conclusion

Summary of project achievements.

- Validated the feasibility of high-performance, strictly Zero-Knowledge enterprise collaboration.
- Proved that advanced cryptography (AES-GCM, Argon2id) does not hinder sub-250ms real-time synchronization constraints.
- Successfully replaced weak password paradigms with highly secure, cryptographic challenge-response authentication.
- Delivered a fully functional, self-hosted architecture guaranteeing digital sovereignty for enterprises.
- Effectively protected extremely confidential workflows from lateral movement, cloud breaches, and insider threats.



Future Scope

The roadmap for continued development:

- **MLS Protocol Integration:** Migrating from Signal's pairwise encryption to the IETF MLS standard for infinitely scalable secure group chats.
- **AI-Assisted Threat Detection:** Implementing local, on-device AI models to detect malicious insider activity without breaking encryption.
- **Native Mobile App Support:** Expanding from desktop (Tauri) to fully native iOS and Android environments.
- **Federated Deployments:** Allowing independent Trustline enterprise servers to securely interoperate and collaborate.
- **Decentralized Identity:** Integrating blockchain-based DID (Decentralized Identifiers) for cross-organizational trust.



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Academic and technical sources

- [1] K. Liang et al., "A Lightweight and Efficient Privacy-Preserving Scheme for Data Security Analysis in Clouds," *IEEE Access*, vol. 8, pp. 100700–100710, 2020.
- [2] A. Smith, "Zero-Knowledge Proofs for Confidential Transactions," *WJARK*, vol. 1, pp. 1–10, 2021.
- [3] M. Cohn-Gordon et al., "A Formal Security Analysis of Signal Messaging Protocol," in *IEEE European Symposium on Security and Privacy*, 2017.
- [4] W3C, "Web Authentication: An API for accessing Public Key Credentials Level 2," 2021.
- [5] N. Preguiça et al., "Conflict-free Replicated Data Types (CRDTs)," 2018.

Thank You!



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Additional academic and technical sources:

- [6] "The Complete Guide to AES 256 GCM Encryption," ResearchGate, 2021.
- [7] "Encryption Algorithms and Their Applications in Cloud Security," arXiv, 2026.
- [8] "A Cloud Data Security Framework Using AES 256 GCM Encryption," ResearchGate, 2021.
- [9] S. K. Singh et al., "Evaluating Argon2 Adoption and Effectiveness in Real-World Scenarios," arXiv, 2025.
- [10] "PostgreSQL Level Security," PostgreSQL Global Development Group.

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Targeted domains for project publication:

- Potential Publications Domains:
- Target Conferences Domains:
- IEEE Symposium on Security and Privacy (S&P).
- ACM Conference on Computer and Communications Security (CCS).
- USENIX Security Symposium.

• Key Contribution Focus: The novel combination of Blind Routing, CRDTs, and Argon2id for scalable E2EE.

Enterprise

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Conclusion and Q&A

- Questions & Discussion
- TRUSTLINE
- Secure • Sovereign

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